LAB #3 – Applications of Array Operations

Lab Objective:

- Applications of mathematical operations on arrays with MATLAB.

Lab Preparation:

Read the course slides and chapter 3 of the textbook.

Lab Procedures:

1. With MATLAB, finish **Chapter 3's problems 7 to 10 in the textbook (section 3.9).** Save all the code of these problems in one file and execute them step by step; and Use ***disp*** to display the number index of each question and use ***pause*** command to generate a short pause between the questions.
2. With MATLAB, finish **Chapter 3's problems 20, 22, 24, 29, 33, and 37** in the textbook (section 3.9). Save all the code of these problems in one file and execute them step by step, Use ***disp*** to display the number index of each question and use ***pause*** command to generate a short pause between the questions.

   Note: The textbook is: Gilat, Amos, Wiley(2017). MATLAB: An Introduction with Applications, 6th Edition.

Lab Demonstration Requirements (40%):

Demonstrate at least 2 questions from procedure 1 and 4 questions from procedure 2 above in the lab computer during the lab time.

After Lab Practice Questions (20%):

Evaluate what will be the values of variables in the table after these MATLAB instructions will be executed (**Do these by hand: Do not use MATLAB!**). Fill in the table below these questions with pencil and paper calculations. Must show all your calculation procedures.

```
n1 = [10 14 8 6]
n2 = [6 4 12 2]'
m = n1' + n2
n = n1 - n2
r = n1 .* n2
s = n1 .* n2'
s1 = dot(n1, n2)
s2 = dot(n1', n2)
```

```
n1 = [100:100:400]
p = n1' + n1'''
q = n1' + n1'
[t, u] = max (n1)
```

```
n1 = [10:20:60]
n2 = 'This is for fun?'
v = length (n2)
w = length (n1)
```

In your report, use a table as the following to display the results:

Variable Name	Variable Value
m	
n	
p	

q	
r	
s	
s1	
s2	
t	
u	
v	
w	

Lab Report Submission Requirement (40%):

1. Show the demonstrations as required on the lab computers and submit Lab report based on the provided templates on the Blackboard. Please post all the screenshots of the inputs and running results of each step on the lab manual.
2. **Submit your source code (the two .m files) of procedure 1 and procedure 2 on the blackboard.**
3. Please submit your report on the blackboard by the due time.